Code: DMM-01 *Paper ID: 13111*

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER –III) END TERM EXAMINATION, APRIL-2024

Marketing Analytics

[Time:03:00 Hours] ***[Maximum Marks: 60]***

Following are the course outcomes of the subject: - Marketing Analytics

CO	Course Outcome (CO)
CO1	Understand the concepts of marketing analytics
CO2	Interpret strategies related to marketing analytics for better understanding of market scenarios.
CO3	Apply concepts and strategies related to marketing analytics for better marketing decision making.

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)

Q.1	**Marks**	**CO**
(A)	**4**	CO – 1
(B)	**4**	CO – 1
(C)	**4**	CO – 1
(D)	**4**	CO – 1
(E)	**4**	CO – 1

SECTION-B

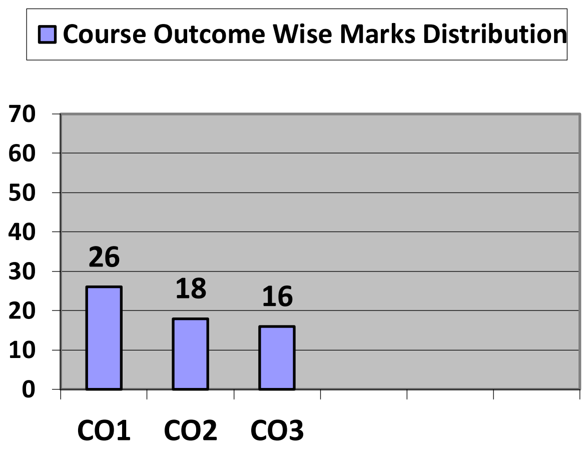
Attempt all questions out of the following: (Long Answer Type Questions)

Q. No	**Marks**	**CO**
2 (A)	6	CO – 1
2 (A)	6	CO – 1
2 (B)	6	CO – 2
2 (A)	6	CO – 2
3 (A)	6	CO – 2
3 (A)	6	CO – 2
3 (B)	6	CO – 2
3 (B)	6	CO – 2

SECTION-C

Attempt all parts of the following: (Caselet / Numerical)

Q.4	**Marks**	**CO**
4 (A)	**8**	CO – 3
4 (B)	**8**	CO – 3



Moderated By **Head- IQAC** **Dean - Examinations**

Printed Pages: 2

Roll No: | | | | | | | | | | |

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[Time: 03:00 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:

(i) ***There are THREE sections in this question paper.***
(ii) ***Attempt questions from each section as per the directions.***
(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: ***(5×4=20)***

(a) Provide a detailed overview about application of analytics in Marketing Decision Making.
(b) State and explain the usage of regression analysis in forecasting.
(c) Discuss the meaning of Customer Lifetime Value. How is it calculated? Explain its application in insurance industry.
(d) What are outliers? Discuss ways in which outliers can be identified. Discuss the various measures of central tendency.
(e) Discuss the application of market basket analysis in designing marketing promotional schemes.

SECTION-B
(Long Answer Type Questions)

Q.2. Attempt all questions of the following ***(4×6=24))***

(a) use ‘cranberrydata.xlsx’. the file contains the data for each quarter in the years 2006 – 2011 detailing the pounds of cranberries sold by a small grocery store. You also see the store’s proce and the price charged by the major competitor. Answer following on the basis of this file.
(i) Create a chart that displays seasonality of cranberry sales and provide analysis
(ii) Ignoring price create a chart that shows whether there is an upward trend in sales and provide analysis

OR

(a) Determine the average sales per quarter, breaking it down based on whether your price has higher or lower than the competitor’s price. Provide the detailed analysis.

(b) The ‘Weather’ worksheet includes monthly average temperature and rainfall in Bloomington, Indian create a combination chart involving a column and line graph with a secondary axis to chart the temperature and rainfall. Also provide analysis of same.

OR

(b) The ‘Survey’ worksheet contains results evaluation a training seminar for sales people. Use a pivot chart to summarize the evaluation data. Also provide the analysis of same.

Q.3. *(a)* Describe the understanding of pricing a product. Also state the relationship between pricing, demand and quantity of sales.

OR

(a) State and explain the price elasticity and inelasticity with suitable examples.

(b) State and explain the linear and power demand curves with suitable examples.

OR

(b) The following is the output of a correlation matrix.

	price	*crime_rate*	*resid_area*	*air_qual*	*room_num*	*age*	*avg_dist*	*teachers*	*poor_prop*	*n_hos_beds*	*n_hot_rooms*	*rainfall*	*parks*	*airport_YES*	*waterbody_Lake*	*waterbody_River*	*waterbody_LakeAndRiver*
price	1																
crime_rate	-0.46653	1															
resid_area	-0.48475	0.660283	1														
air_qual	-0.4293	0.707587	0.763651	1													
room_num	0.696304	-0.28878	-0.39168	-0.30219	1												
age	-0.378	0.559591	0.644779	0.73147	-0.24026	1											
avg_dist	0.249289	-0.58637	-0.70802	-0.76925	0.205241	-0.74791	1										
teachers	0.505655	-0.39005	-0.38325	-0.18893	0.355501	-0.26152	0.232452	1									
poor_prop	-0.74084	0.60897	0.6038	0.590879	-0.61381	0.602339	-0.49697	-0.37404	1								
n_hos_beds	0.10888	-0.00409	0.005799	-0.04955	0.032009	-0.02101	-0.02787	-0.00806	-0.06601	1							
n_hot_rooms	0.017007	0.056569	-0.00376	0.007238	0.014583	0.013918	-0.0207	-0.03701	0.017035	-0.00313	1						
rainfall	-0.0472	0.082151	0.055845	0.091956	-0.06472	0.074684	-0.03728	-0.04593	0.061444	0.058596	0.014868	1					
parks	-0.39157	0.638951	0.707635	0.915544	-0.28282	0.67385	-0.70792	-0.187	0.55231	-0.07127	0.023756	0.078278	1				
airport_YES	0.182867	-0.13449	-0.1154	-0.0739	0.163774	0.005101	0.021402	0.069437	-0.09505	-0.00637	-0.05534	-0.01317	-0.0525	1			
waterbody_Lake	0.036233	-0.02539	-0.02659	-0.04639	-0.0042	0.003452	0.03489	0.048717	0.003197	0.042278	0.037925	-0.01617	-0.03499	0.035491	1		
waterbody_River	0.071751	-0.0601	-0.09898	-0.03777	0.046251	-0.08861	0.032247	0.094256	-0.109	-0.07415	-0.0641	-0.03702	-0.04886	0.017341	-0.36656	1	
waterbody_LakeAndRiver	-0.0375	0.009076	0.051649	0.013849	0.010554	-0.00435	-0.02132	-0.04698	0.02062	0.059482	0.014754	0.109234	0.013265	-0.07034	-0.19675	-0.30409	1

SECTION-C

Q.4. Attempt all parts of the following: ***(2×8=16)***

4(a) A prescription drug is purchased in the United States and sold internationally. Each unit of drug costs $60 to purchase. In the German market you sell the drug for 150 Euros per unit. The currency exchange is $0.667 per euro. Current demand for the drug is 100 units, and the estimated elasticity is 2.5. Assuming a linear demand curve, determine the appropriate sales price (in euros) for drug.

4(b) You want to determine the correct price for a new weekly magazine. The variable cost of printing and distributing a copy of the magazine is $0.50. You are thinking of charging from $0.50 through $1.30 per copy. The estimated weekly sales of the magazine are shown in the following table. What price should you charge for the magazine.

Price	Demand (in millions)
$0.50	2
$0.90	1.2
$1.30	3